

Haobin Chen

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Add.: No.777, Xinye Street, Fanyu District, Guangzhou, Guangdong, China

EDUCATION

South China University of Technology (SCUT)

Master of Engineering in Mechanical Engineering

Guangzhou, China

Sept.2024 – Jun.2027

Recommended Admission (Exempt from National Entrance Examination).

• **GPA: 85/100**

• **Awards:**2024 & 2025 & 2026 Postgraduate Scholarships

SouthWest Petroleum University (SWPU)

Bachelor of Engineering in Mechanical Design, Manufacturing and Automation

Chengdu, China

Sept.2020 – Jun.2024

• **GPA:** 88.7/100; **Ranking:** 4/94

• **Awards:** 2022 Wewodon Scholarship(1%); 2022 & 2023 First-Class Scholarship (3%)

PUBLICATIONS&PATENTS

- **H. Chen**, Y. Chen, L. Xie., "PMG-RL for Stable Multi-Speed Walking Control of a Human Musculoskeletal Model," 2026 *IEEE 3rd International Conference on Energy and Electrical Engineering*, Jun. 2026. (Accepted & Indexed by EI)
- **H. Chen**, F. Li, L. Xie et al., "NORM-FoG: Normal-Gait-Oriented Residual Modeling for IMU-Based Freezing-of-Gait Detection," *Expert Systems with Applications*. (Under Review)
- L. Chen, **H. Chen**, L. Xie et al., "FrontalSleepNet: Multiscale Context-Aware Sleep Staging from Information-Limited Frontal EEG," *IEEE Journal of Biomedical and Health Informatics*. (Under Review)
- **H. Chen**, Y. Chen, L. Xie et al., "CPG-Guided Reinforcement Learning for Hip Assistance Strategy in a Musculoskeletal Environment," *IEEE Robotics and Automation Letters*. (In Preparation)
- H. Wang, **H. Chen**, et al., "A Device and Method for Testing Phase-Change Thermal Storage Material Performance," China Invention Patent ZL202311331955.0, Granted Oct. 2025.
- H. Wang, **H. Chen**, et al., "A Phase-Change Material Thermal Storage Performance Testing Device," China Utility Model Patent ZL202322761347.5, Granted May 2024.
- Y. Wang, Y. Mao, **H. Chen**, et al., "A Novel Infrared Non-Destructive Fruit Sugar Content Measurement Device," China Utility Model Patent ZL202223027103.6, Granted Jun. 2023.
- H. Wang, H. Shi, **H. Chen**, et al., "A Machine-Vision-Based Component Sorting Device," China Utility Model Patent ZL202120585669.7, Granted Nov. 2021.

RESEARCH EXPERIENCE

Adaptive Hip Exoskeleton System for Freezing-of-Gait in Parkinson's Disease

Guangzhou, China

Guangdong S&T Program (Grant No. 2026B0101110003)

Advisor: Dr. Longhan Xie

Jul.2025 – Present

- Designed and developed a portable hip exoskeleton by integrating mechanical structure, sensors, embedded hardware, and real-time control algorithms into a functional assistive system;
- Proposed a framework for freezing-of-gait (FOG) detection that models normal-gait patterns and identifies abnormal deviations, mitigating the class imbalance and inter-subject variability (**1 publication**);
- Designed a multi-site IMU data collection study with 9 patients with Parkinson's disease (PD), establishing an annotated FoG dataset for gait-assistance and detection research;
- Conducted human-subject experiments to evaluate the assistive and clinical effects of the hip exoskeleton, assessing its effectiveness in reducing FoG episodes and improving gait performance in patients with PD (**Planned Evaluation**).

EEG-fNIRS Multimodal Brain-Controlled Rehabilitation Robot

Guangzhou, China

Natural Science Foundation of Guangdong Province (Grant No.2024A1515011190)

Advisor: Dr. Longhan Xie

Sept.2024 – Jun.2025

- Developed a prefrontal low-channel multimodal acquisition framework to address the limitations of traditional full-scalp EEG caps (e.g., time-consuming setup and poor comfort);

- Led hardware development of the multimodal brain-computer interface system, designing differential transmission circuits and common-mode filtering to resolve cross-modal interference and temporal inconsistency during synchronized EEG-fNIRS acquisition;
- Determined spatially isolated FPC layouts and time-division multiplexing driving strategies through standard signal validation experiments, effectively suppressing inter-modal crosstalk;
- Achieved experimental performance metrics including time constant error < 2%, equivalent input noise < 3 μ V, and power spectrum amplitude-frequency deviation < 2%;
- Contributed to system commercialization; the project has received medical device registration approval;
- This work contributed to one accepted conference paper and one manuscript in preparation.

Dual-Manipulator Cooperative System for Drilling Derrick Pin Assembly and Disassembly **Chengdu, China**
PetroChina Project

Advisor: Teng Hu Sep.2023 – Dec.2024

- Compiled technical specifications of subsystems and performed quantitative analyses to support feasibility assessment and equipment selection, including deriving the relationship between robot payload capacity and workspace radius;
- Proposed a multi-robot cooperative architecture integrating two collaborative robots and one hydraulic robot, enabling coordinated force–position operation and load sharing for automated pin disassembly and assembly;
- Designed a robotic end-effector for derrick truss pin disassembly and assembly, incorporating gear transmission and structural force-transfer mechanism to reduce robot-end loads and enable smooth operation under high reaction forces.

Phase-Change Material Property Testing Instrument Design **Chengdu, China**

Advisor: Haohan Wang Apr.2023 – Jun.2024

- Addressed the high cost and low throughput of traditional DSC/TGA methods by performing multi-physics coupling optimization on the tester structure using Finite Element Analysis (FEA) based on latent heat and energy density models;
- Designed and implemented a low-cost performance tester, completing heat transfer process modeling and experimental validation;
- Reduced per-test cost to under ¥100 (approx. 45% lower than traditional DSC/TGA) and compressed the operation to 5 steps, enabling high-throughput thermal storage performance evaluation;

Apple Sugar Content Distribution Monitoring and Management System **Chengdu, China**

Advisor: Yunxu Zhou Jun.2022 – Oct.2023

- Designed the optomechanical structure and manufactured a prototype based on Near-Infrared (NIR) spectroscopy; optimized the optical path anti-interference structure to enhance measurement consistency;
- Built a non-destructive sugar content detection platform achieving 96.8% prediction accuracy;
- Integrated mobile-end data acquisition and visualization modules for real-time quality monitoring and classification management;

PROFESSIONAL EXPERIENCE

Lychee Intelligent Technology Co., Ltd. **Guangzhou, China**

R&D Engineer Jun.2025 – Dec.2025

- Developed hip-assist exoskeleton systems for rehabilitation and augmentation applications; responsible for full-pipeline R&D from **structural design, control software development, and prototype iteration to small-batch production** and certification;
- Iterated the product for 5+ rounds, optimizing assistance performance and system stability and achieving small-batch production of 100+ units;
- Facilitated 3C certification and medical device approval, completing the transition from prototype to engineered product;

Lincseek Technology Co., Ltd. **Chengdu, China**

R&D Engineer Nov.2023 – Jun.2024

- Designed and fabricated a six-DOF serial robot prototype; independently developed forward/inverse kinematics control algorithms and host PC-based graphical user interface;
- Ranked 1st in undergraduate thesis defense and received University-level Undergraduate Thesis Innovation Award (1 recipient across the entire major);

TEACHING

Engineering Training Center, Southwest Petroleum University

Chengdu, China

Teaching Assistant

Sept.2021 – Jun.2023

- Assisted in laboratory management and experimental instruction for the Engineering Cognition Practice course;
- Supported robotics lab personnel management and competition technical assistance;

HONORS

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- National Gold Award, China International College Students' Innovation Competition (2025)
 - Grand Prize, 18th Guangdong Extracurricular Academic Science and Technology Competition (2025)
 - First Prize, 16th National Undergraduate Advanced Mapping Technology and Product Information Modeling Innovation Competition (Additive Manufacturing Track, 2023)
 - National First Prize, National Intelligent Robot Creative Design Competition for University Students (2023)

TECHNICAL SKILLS

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- **Robotics & Simulation:** Forward/Inverse Kinematics & Dynamics, Motion planning, ROS, MuJoCo (Sim-to-Real, RL Integration)
 - **Hardware & EDA:** CAD/CAE/CAM (e.g., SolidWorks, UG, Ansys), EDA (e.g., EasyEDA)
 - **Programming:** Python, C, C++, Matlab